

SI140A: Probability & Statistics for EECS

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Outline

- 1 Teaching Team
- 2 Course Information
- 3 Probability & Statistics

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The Dark Forest in Random Kingdom



Life is Random



**Life was like a box
of chocolates. You
never know what
you're gonna get.**



Forrest Gump [1994]
www.geckoandfly.com

Even Great Scientists Failed

- Isaac Newton: “I can calculate the motion of heavenly bodies, but not the madness of people.”



Even Great Scientists Failed

- Albert Einstein: “God does not play dice with the universe.”



Sometimes Mysterious

- Schrodinger's Cat: “遇事不决，量子力学；解释不通，穿越时空.”



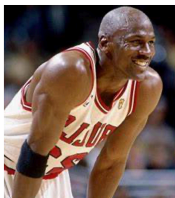
Sometimes Controversial

- Mendel-Fisher Controversy: Cherry Picking?



Sometimes Controversial

Case 3: Who is the greatest NBA player?



VS



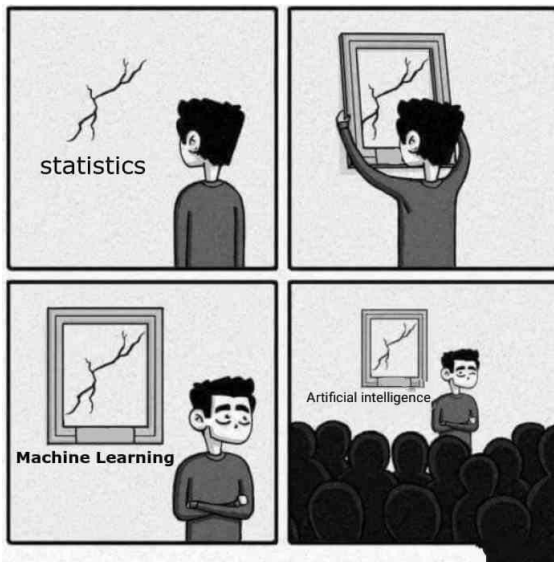
篮球得分	乔丹	詹姆斯
二分之一球	$12192/24537 = 49.7\%$	$12424/24654 = 50.4\%$
三分球	$581/1778 = 32.7\%$	$1860/5409 = 34.4\%$
总计	$12773/26315 = 48.5\%$	$14284/30063 = 47.5\%$

Sometimes Reinventing the Wheel

Reduction of random noise has many names in different fields.

- Terminology in EECS: signal processing (filter)
- Terminology in Probability & Statistics: estimation
- Terminology in ML & AI: prediction

Sometimes Reinventing the Wheel



You Need Shepherd!



TAs and Professors

- TA Shangshang Wang: 王上上
- TA Yuhan Wang: 王宇翰
- TA Bowen Xu: 徐博文
- Professor Dingzhu Wen: 文鼎柱
- Professor Ziyu Shao: 邵子瑜

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<https://faculty.sist.shanghaitech.edu.cn/faculty/shaozy>

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Time & Venue

- **Time:** every Tuesday (10:15-11:55am) and Thursday (10:15-11:55am)
- **Location:** Room 302, Teaching Center
- **Course Forum in Piazza:**
https:
[//piazza.com/shanghaitech.edu.cn/spring2023/si140a](https://piazza.com/shanghaitech.edu.cn/spring2023/si140a)

Course Grade

- *Homework* 20%: 5 or 6 problems per week
- *Behavior* 5%: miscellaneous behaviors
- *Final Project* 15%: due after final exam
- *Midterm Exam* 30%
- *Final Exam* 30%
- *Overall Winners*: lunch
 - ▶ Gauss Award
 - ▶ Markov Award
 - ▶ Bernoulli Award
 - ▶ Laplace Award
 - ▶ Fisher Award

Homework Policy

- Submit homework through Gradescope.
- Write the homework with LaTeX and output it as a PDF file.
- Write the programming assignment with Python and submit source codes.
- **Later** Homework receives no credit.
- You are allowed to discuss with others and use any references, but if you do so please list your collaborators and cite your references for each question.
- Not writing your own solutions or not listing your collaborators or not citing your references may be considered plagiarism.

Project Policy

- Project can be done by a single student or a team with two students.
- Teams with three and more students are NOT allowed.
- Your team is required to use Python for the programming part.
- Your team needs to submit both the Python code and a report (in PDF format) including your simulation results, analysis, discussions, tables, figures, etc.
- **Later** Project receives no credit.

LaTeX Setup

- LaTeX Package (Unix, Windows): TeXLive (<https://tug.org/texlive/>)
- LaTeX Package (MacOs): MacTeX (<http://tug.org/mactex>)
- Popular LaTeX Editors
 - ▶ Texpad (MacOs): <https://www.texpad.com>
 - ▶ TeXstudio (Windows,MacOs): <http://www.texstudio.org>
 - ▶ LyX(Windows,MacOs): <https://www.lyx.org>
 - ▶ WinEdt (Windows): <http://www.winedt.com/>
- Online LaTeX Editor: Overleaf (<http://overleaf.com/>)

LaTeX Tutorials

- Wiki LaTeX Book: <http://en.wikibooks.org/wiki/LaTeX>
- Collection of links for LaTeX: <https://www.tug.org/begin.html>
- Forum: <https://tex.stackexchange.com/>

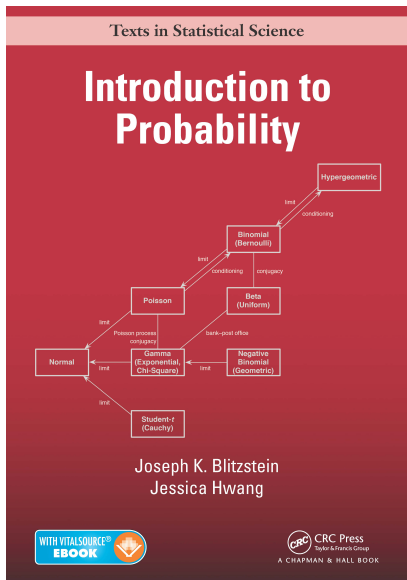
R: Optional

- Setup: <https://www.r-project.org/>
- Editor: <https://www.rstudio.com/>
- Data Visualization in R: <http://ggplot2.org/>
- Reference 1: *R in Action: Data analysis and graphics with R*, Robert I. Kabacoff, Manning Publications, 2015.
- Reference 2: *R for Data Science*, Hadley Wickham & Garrett Grolemund, O'Reilly Media, 2016.

Textbooks: Required

- *Introduction to Probability*, Joseph K. Blitzstein & Jessica Hwang, Chapman & Hall/CRC, 2014. (**BH**)
- *Introduction to Probability (2nd Edition)*, Dimitris P. Bertsekas & John N. Tsitsiklis, Athena Scientific, 2008. (**BT**)

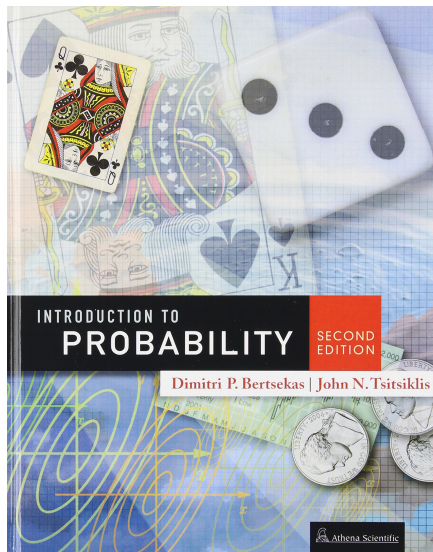
Required Textbook: BH



Joseph K. Blitzstein &
Jessica Hwang

- Introduction to Probability
- Chapman & Hall/CRC, 2014.

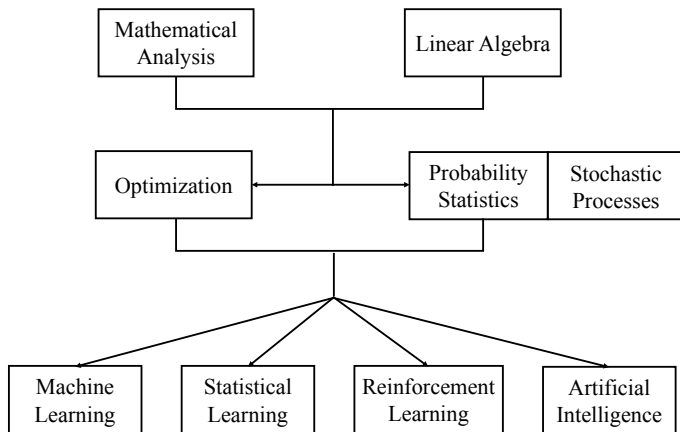
Required Textbook: BT



Dimitris P. Bertsekas &
John N. Tsitsiklis

- Introduction to Probability (2nd Edition)
- Athena Scientific, 2008.

The Role of This Course



Course Objectives

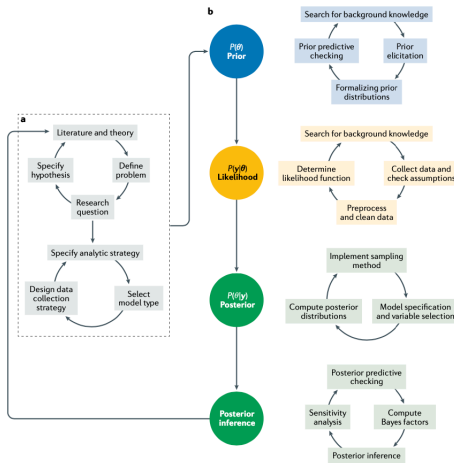
- Integration of Probability, Statistics, Data Science, Random Algorithm
- **Conversation between mathematics and engineering:** theory, model, algorithm, practice
 - ▶ **Solid foundation of probability & statistics:** modeling, analysis, and theoretical proof
 - ▶ **Computational thinking:** algorithmic perspective and programming practice
 - ▶ **Application mentality:** many examples from EECS

Course Focus

- **Motivation:** In the ocean of mathematical definitions, theorems, and equations, why should we spend our time on this particular topic but not another?
- **Intuition:** When going through the derivations, is there a physical intuition beyond those equations?
- **Implication:** After we have learned a topic, what new problems can we solve?

Theme

- One Core: Bayesian Inference
- Two Basic Tools: Conditional Probability & Conditional Expectation



Basic Contents: Traditional Perspective

- Probability and Counting
- Conditional Probability
- Random Variables and Distributions
- Expectation
- Continuous Random Variables
- Joint Distributions
- Transformations
- Conditional Expectation
- Inequalities & Limit Theorems & Monte Carlo
- Classical Statistical Inference
- Bayesian Statistical Inference
- Markov Chain (optional)

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Probability vs. Statistics

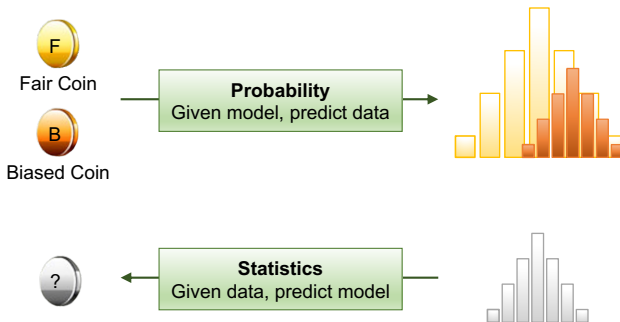


Statistics: Given the information in your hand, what is in the pail?



Probability: Given the information in the pail, what is in your hand?

Probability vs. Statistics



Probability vs. Statistics

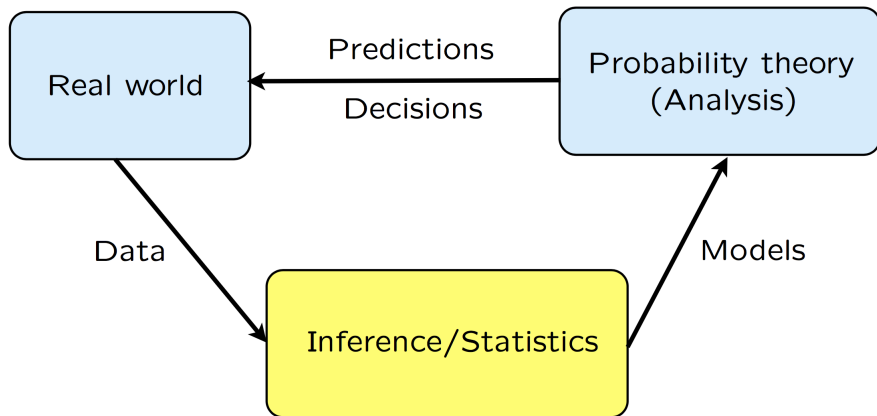
- **Focus of Probability:** given a data generating process (model), what are the properties of the outcomes?
- **Focus of Statistics:** the inverse of probability, i.e., given the outcomes, what can we say about the process that generated the data (model)?

随机非随意
概率破玄机
无序隐有序
统计解谜离

词以境界为最上。有境界自成高格，
自有名句。有造境，有写境，此理想
与现实二派之所分。

人间词话 - 王国维

A Framework to Quantify Uncertainty



Let the Adventure Begin!

